

UNIT 1 – ELECTRIC CHARGE, FORCE, AND FIELD

1.2 Electric Field of Point Charges**Answer key**

Placeholder worksheet for the site prototype. Free for classroom use.

1. Find the magnitude of the electric field 0.050 m from a point charge of +8.0 nC.

$$E = \frac{kq}{r^2} = \frac{(8.99 \times 10^9)(8.0 \times 10^{-9})}{(0.050)^2} = 2.9 \times 10^4 \text{ N/C, directed away from the charge.}$$

2. A charge $+q$ sits at $x = 0$ and a charge $+4q$ sits at $x = d$. Where on the x -axis is the net electric field zero?

Only between the charges can the fields cancel. Set $\frac{kq}{x^2} = \frac{4kq}{(d-x)^2}$, so $(d-x)^2 = 4x^2$ and $d-x = 2x$, giving $x = d/3$.

3. An electron is released from rest in a uniform field of $2.0 \times 10^3 \text{ N/C}$. Find its acceleration (magnitude and direction relative to $\vec{E}$).

$$a = \frac{eE}{m_e} = \frac{(1.60 \times 10^{-19})(2.0 \times 10^3)}{9.11 \times 10^{-31}} = 3.5 \times 10^{14} \text{ m/s}^2, \text{ opposite to } \vec{E} \text{ because the electron is negative.}$$